

MSME

INNOVATION IN ADOPTION OF INDUSTRY 4.0 AND 5.0 IN MSME ECOSYSTEM

2. DETAILS OF IDEA:

2.1 TITLE PROPOSED:

DIGITAL TWIN FOR MICRO-MANUFACTURING

2.2 WHETHER THE IDEA INVOLVES USE OF EXISTING INTELLECTUAL PROPERTY OR NOT, GIVE BRIEF DETAIL THERE OF

Idea does not rely on proprietary IP. It utilizes open-source digital twin platforms, low-cost IOT kits, and cloud-based monitoring solutions. All technologies used are publicly available or can be developed with open collaboration. For example, we plan to use tools like Eclipse Ditto, FIWARE, or Node-RED, along with affordable hardware such as ESP32 boards. These are widely used in the tech community and are known for being budget-friendly, especially for small manufacturers. The system is designed to be modular and flexible, so manufacturers can choose what they need — like temperature sensors, machine monitoring tools, or energy trackers — and easily plug them in. This helps even the smallest units monitor and improve their operations using digital twin technology without big investments.

2.3 UNIQUENESS OF THE INNOVATION IDEA:

This innovation brings Digital Twin Technology, typically used by large enterprises, into the MSME sector by offering a cost-effective, scalable solution. It leverages open-source tools, affordable hardware, and community partnerships to enable real-time machine monitoring, predictive maintenance, and efficiency optimization—something traditionally unavailable to smaller firms due to high costs and technical barriers. The system uses easy-to-set-up IOT kits and free software platforms, so even small manufacturers can start with minimal investment. Components like ESP32 are used to collect machine data, which is then visualized through simple dashboards using open-source tools like Node-RED or Eclipse Ditto. The goal is to make advanced tech practical and accessible—not just for factories with big budgets, but for small workshops too. This helps create a more inclusive, competitive, and future-ready manufacturing ecosystem.

2.4 CONCEPT & OBJECTIVE:

CONCEPT

The concept is to use **Digital Twin technology** to create **virtual replicas of machines and manufacturing processes** in micro and small manufacturing units (MSMEs). These digital twins mirror real-time data from the physical systems, allowing businesses to **monitor performance, track operations, and analyze machine behavior** in a digital environment. The solution integrates **open-source tools, affordable IoT hardware, and cloud-based dashboards** to ensure that even the smallest firms can benefit from insights traditionally reserved for large enterprises.

OBJECTIVE

The objective is to **bridge the digital divide** in the MSME sector by making advanced technologies like Digital Twin accessible and practical. By implementing this solution, MSMEs will be able to:

- **Monitor real-time machine performance**
- **Predict and prevent equipment failures**
- **Reduce downtime**
- **Improve overall production flow**

Ultimately, the goal is to deliver a **cost-effective, easy-to-adopt** system that enhances **productivity**, supports **sustainability**, and improves **competitiveness** in the evolving **digital economy**.

2.5 POTENTIAL AREAS OF APPLICATION IN INDUSTRY/MARKET :

Manufacturing units (Textile, Auto-parts, Food processing) Digital Twins can track real-time production data, machine health, and resource use. This helps optimize workflows, reduce waste, and improve product quality across various sectors.

Packaging and Assembly operations By simulating and monitoring packaging and assembly lines, businesses can detect inefficiencies, predict breakdowns, and ensure smoother operations with minimal human intervention.

Small-scale Machine shops Machine shops can use Digital Twins to visualize tool wear, plan maintenance, and increase machine uptime. This results in better scheduling and reduced operational costs.

Maintenance and Equipment Monitoring Services Service providers can remotely monitor client equipment, predict failures, and offer proactive maintenance. This boosts service quality while reducing unplanned outages.

Training and Simulation environments Digital Twins offer safe, realistic simulations for operator training. Trainees can interact with virtual machines to understand processes without disrupting actual production.

2.6 THE MARKET DATA FOR THE POTENTIAL IDEA/ INNOVATION:

The adoption of **Digital Twin Technology** in the MSME sector is part of the broader push toward digital transformation in manufacturing, driven by affordability, efficiency demands, and global competitiveness.

1. Market Size and Growth:

India has over **63 million MSMEs**, contributing **30% of the national GDP** and employing more than **110 million people**. However, most lack access to advanced technologies.

Globally, the **Digital Twin market is projected to grow at a CAGR of 40%**, expected to **exceed USD 100 billion by 2030**, indicating massive opportunity for industrial digitization.

2. Key Growth Drivers:

- **Technology Accessibility:** Open-source platforms and low-cost IoT hardware are making Digital Twin solutions viable for small businesses.

- **Productivity & Efficiency Needs:** MSMEs need solutions to reduce downtime and improve process control. Digital Twins can boost **efficiency by 15–30%** and cut **downtime by 25–40%**.
- **Government Support:** Policies like *Digital MSME* and *Make in India* are promoting tech adoption in the sector.
- **Sustainability Goals:** Better monitoring helps reduce energy use and waste, aligning with green manufacturing goals.

3. Regional Insights:

India, as one of the largest MSME hubs, is well-positioned to lead in affordable digital manufacturing. Adoption is also rising in Southeast Asia and Africa, where small industries dominate production.

4. Competitive Landscape:

Startups and solution providers focusing on **open-source, modular, and low-cost Digital Twin frameworks** are emerging rapidly. Collaboration between tech developers, academic institutions, and industry bodies is key to scaling adoption in MSMEs.

5. Potential Customers:

Primary users include **small manufacturing units** in textile, auto-parts, food processing, machine shops, and packaging. Secondary users include **training institutes, equipment monitoring services, and government-backed technology clusters**.

2.7 NAME OF THE MENTOR:

NAME: DR.J.SASIDEVI

COLLEGE: K.RAMAKRISHNAN COLLEGE OF ENGINEERING

DEPARTMENT: COMPUTER SCIENCE AND BUSINESS SYSTEM

2.8 EXPERIENCE AND QUALIFICATION OF MENTOR:

EXPERIENCE: 23 YEARS TEACHING EXPERIENCE

QUALIFICATION: M.E.,Ph.D

2.9 CONTACT DETAILS OF MENTORS:

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2.10 CURRENT DEVELOPMENT STATUS OF INNOVATION:

The idea is currently in the **conceptual design and validation stage**. Initial simulations have been completed, and planning for the **IoT integration and cloud-based monitoring setup** is underway. We're also identifying the most cost-effective hardware and open-source tools to ensure accessibility for MSMEs. Early feedback from potential users is being considered to refine the design and use-case scenarios..

2.11 EXPECTED TIME OF COMPLETION OF IDEA:

The project timeline is estimated to be **6 to 8 months**, which includes **prototyping, testing, and field deployment**.

The first phase will focus on building and refining the prototype using selected IoT and digital twin tools. This will be followed by lab testing, real-world trials in small manufacturing units, and final adjustments based on user feedback to ensure the solution is practical, reliable, and easy to adopt.

3. FINANCIAL REQUIREMENTS:

3.1 Please give Total idea/project cost (Rs in lakh):

- Total idea/project cost: ₹12 lakh
- Total GOI (Govt of India) Assistance required: ₹9 lakh
- Total Incubatee share: 0

Particular/Item (As per the scheme guidelines)	Amount GOI (Govt of India) assistance (Rs. In lakh)	Incubatee share (Rs. In lakh)
Technology related Expenditure towards machine usage charges etc., Electricity charges, Procurement of raw material, testing/Calibration charges, other charges essential for development of idea Max (10.00) lakh.	7 LAKH	0
Charges for mentor/handholding supporting team Max (3.00) lakh.	1.5 LAKH	0
Travelling Expenses or any other item not covered as above may be allowed as per need for development of the idea Max (2.00) lakh.	1 LAKH	0

4.NAME OF THE TEAM MEMBERS:

NAME	AADHAR NUMBER
REENA S	8840 1768 5102
SHREE HARINI S	3195 1825 4735
SOUNDARYA AL	7204 6484 0685

5. SUMMARY OF THE IDEA:

This project aims to bring **Digital Twin technology** to India's **MSME sector** by developing a low-cost, open-source solution for real-time machine monitoring and predictive maintenance. Using affordable IoT kits, cloud platforms, and open-source tools, the system creates virtual replicas of machines to track performance, reduce downtime, and improve efficiency. The solution is scalable, accessible, and easy to adopt—bridging the digital gap for small manufacturers and enabling smarter, more sustainable operations.

6 (A) IS IT A NEW CONCEPT?

Yes

(B) PRIOR ART ON THE CONCEPT,

Digital Twin technology is widely used in large industries like aerospace and automotive for system simulation and monitoring. Prior solutions are expensive and complex, limiting MSME adoption. Our approach uses open-source tools and low-cost IoT for broader accessibility.

7. PROBLEM STATEMENT :

Despite their vital role in the economy, most **MSMEs lack access to advanced technologies** like Digital Twins due to **high costs, complexity, and limited technical expertise**. This results in **poor machine visibility, unplanned downtime, and inefficient production**. There is a pressing need for an **affordable, easy-to-adopt solution** that enables small manufacturing units to monitor, predict, and optimize operations in real time.

8. BACKGROUND FOR GETTING THE IDEA?

a. Who is it for?

This solution is designed for **micro, small, and medium enterprises (MSMEs)**, particularly in sectors like **textiles, auto-parts, food processing, packaging, and small machine shops**. It also serves **training institutions and maintenance service providers** looking to adopt affordable Industry 4.0 tools.

b. What will it do?

It will create **Digital Twins (virtual replicas)** of machines and processes, enabling **real-time monitoring, predictive maintenance, and data-driven decision-making**. This helps reduce downtime, improve productivity, and optimize production flows in small-scale manufacturing setups.

c. Any unique features?

The solution stands out for its **low-cost, open-source approach**, making it easy to adopt without heavy investments. It uses affordable IoT devices and cloud platforms, requires minimal technical expertise, and can be **scaled or customized** based on specific MSME needs, promoting inclusive digital transformation.

9. HOW SIMPLE OR COMPLEX WILL THE IDEA’S EXECUTION OR IMPLEMENTATION BE?

The idea is **moderately simple to implement** using open-source platforms, affordable IoT kits, and cloud services. Major tasks include sensor integration, data mapping, and dashboard setup. Risks include limited tech know-how among MSMEs and occasional network issues, which can be managed through training and support.

10. HOW SOON COULD THE IDEA BE PUT INTO OPERATION?

Prototype will be ready in **6–8 months**, currently at **TRL 3–4** (concept validated, initial simulations done). Early deployment in MSMEs will follow testing and refinement.

11. HOW MUCH INVESTMENT WOULD YOU NEED FOR PROTOTYPING OF THE IDEA?

Item	Quantity	Estimated Cost (₹)
IoT Boards & Microcontrollers	5	25,000
Sensors (Temp, Vibration, Current)	15	40,000
Cloud Services & Dashboard Setup	1 (annual)	50,000
Software Development & Integration	-	1,00,000
Field Testing & Pilot Deployment	2–3 sites	75,000

| **Total Estimated Cost** | | **₹2,90,000** |

12(A). HOW DO YOU INTEND TO PROTECT YOUR IDEA (IP)?

The solution is based on **open-source frameworks**, but **unique workflows, system design, and deployment model** may be eligible for a **process or utility patent**. IP will be protected through formal filings as the solution is finalized.

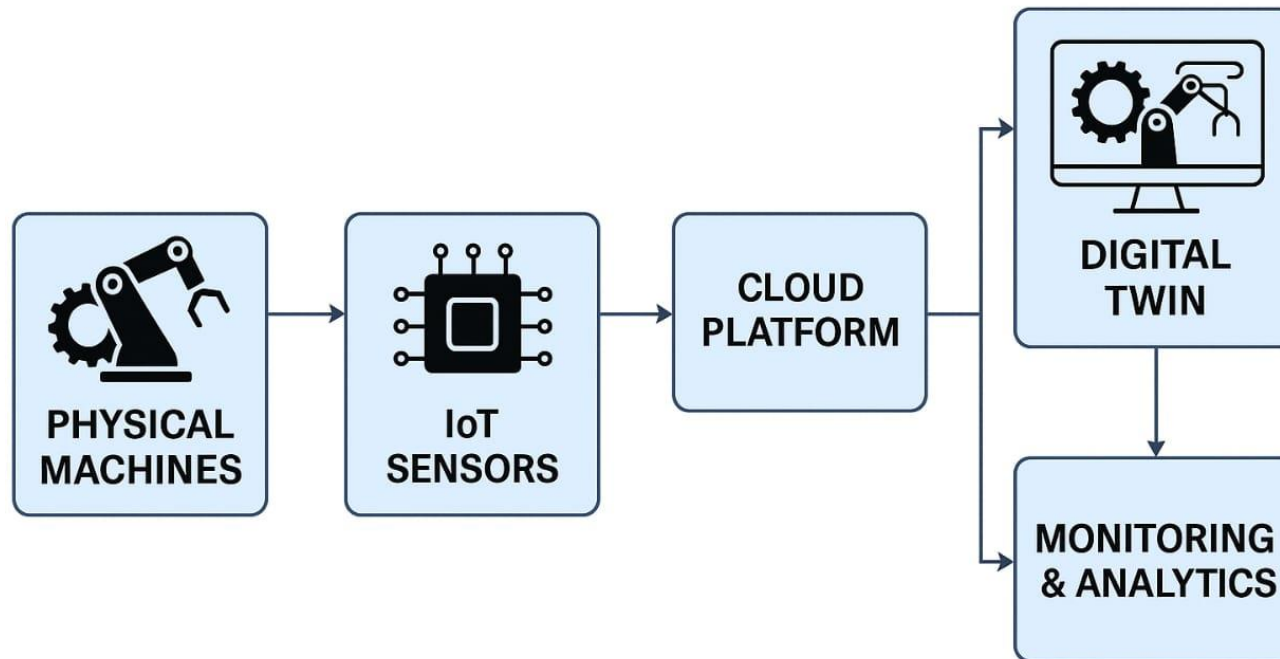
(B). RELATED BACKGROUND:

Digital Twin technology is widely applied in large industries like aerospace, auto, and energy. Companies like Siemens and GE use high-cost proprietary platforms. However, no scalable, low-cost solution exists for MSMEs using open-source tools and modular IoT integration—our focus fills that gap.

13. HOW IS THIS PROJECT MADE AND USED?

The system uses **IoT sensors** (for temperature, vibration, etc.) connected to **microcontrollers** (e.g., ESP32). Data is transmitted to a **cloud dashboard** for real-time monitoring and alerts. The digital twin simulates machine behavior, predicts faults, and helps MSME operators optimize operations easily.

BLOCK DIAGRAM:



MICROMANUFACTURING

